

These are a collection of notes on Lie groups taken from Richard Borchers' videos.

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1 Introduction

A *Lie group* is a group G that is also a topological space so that the composition and inverse operations are continuous. In the dimension 0 case, Lie groups are the discrete groups. Classifying all the discrete groups is a massive challenge, since the classification of just the finite simple groups was monumental. However, we can make the following useful observation. Say G is a dimension 0 Lie group. Then we get an exact sequence $1 \rightarrow G_0 \rightarrow G \rightarrow G/G_0 \rightarrow 1$. Here, G_0 is a connected component of G and the quotient G/G_0 is a discrete subgroup. As an example, take $G = \mathbb{R}^\times$ with group operation being multiplication. (Certainly this is a Lie group, as composition and inverse operations are essentially multiplications by scalars, which are continuous.) Then, we can take the connected component G_0 to be $\mathbb{R}_{\geq 0}$ and so $G/G_0 = \{+, -\}$, which is discrete. In the dimension 1 case, we have 3 examples, namely $(\mathbb{R}, +)$, $(\mathbb{R}^\times, \times)$, and the unit circle group $(S^1 \subset \mathbb{C}^\times, \times)$. The exponential map $\exp$ gives a homomorphism of $(\mathbb{R}, +)$ to $(\mathbb{R}^\times, \times)$. We also get a homomorphism $(\mathbb{R}, +) \rightarrow (S^1, \times)$ given by the map $x \mapsto \exp(2\pi ix)$. This is called a *local isomorphism* in the sense that the map is an isomorphism when it maps a sufficiently small neighborhood of the origin in $(\mathbb{R}, +)$ to a sufficiently small neighborhood of $1 \in S^1$ (we get the restriction on the size of these neighborhoods due to branch cuts in the complex plane).

Proposition. Any connected 1-dimensional Lie group is isomorphic to $(\mathbb{R}, +)$ or S^1 .

In the dimension 2 case, there are four Lie groups. The abelian ones are $\mathbb{R}^1 \times \mathbb{R}^1$ (the standard 2-dimensional real plane), $\mathbb{R}^1 \times S^1$ (an infinite cylinder), and $S^1 \times S^1$ (the torus T^2).

Proposition. Any abelian connected Lie group is isomorphic to $\mathbb{R}^m \times (S^1)^n$, which is isomorphic to $\mathbb{R}^{m+n}$ modulo a discrete subgroup.

The first nonabelian Lie group is the so-called “ $ax + b$ ” group (denote by G), which is the group of linear maps $\phi : \mathbb{R} \rightarrow \mathbb{R}$ so that $\phi : x \mapsto ax + b$. We can think of elements of $ax + b$ group as matrices of the form

$$\begin{pmatrix} a & b \\ 0 & 1 \end{pmatrix}$$

and we can view an input $x \in \mathbb{R}$ as a column vector $\begin{pmatrix} x \\ 1 \end{pmatrix}$. This group is *solvable* in the sense that there exists a chain of subgroups $1 \subset G_0 \subset G_1 \subset \cdots \subset G_n = G$ (where G is the $ax + b$ group) so that $G_i \subset G_{i+1}$ is normal and G_i/G_{i-1} is abelian. Explicitly, the chain we get for this group is given by the setting G_0 to be the subgroup consisting of solely the identity matrix, G_1 to be the group consisting of matrices of the form

$$\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$$

and G_2 to be G . Then it follows that each quotient G_i/G_{i-1} is abelian and each $G_i \subset G_{i+1}$ is a normal subgroup.

In the dimension 3 case, there are several key examples. We have $SL_2(\mathbb{R})$, which is the group of invertible 2×2 matrices of real entries of determinant 1. We also can mod $SL_2(\mathbb{R})$ by plus or minus the identity matrix to get the group $PSL_2(\mathbb{R})$, which is the group of automorphisms of the upper-half plane $\mathbb{H} \subset \mathbb{C}$ of the form

$$\tau \in \mathbb{H} \mapsto \begin{pmatrix} a & b \\ c & d \end{pmatrix} \tau = \frac{a\tau + b}{c\tau + d}$$

We also have the group of unit quaternions, S^3 , which is the collection of elements $a + bi + cj + dk$ so that $ij = k = -ji, jk = i = -kj, ki = j = -ik$ with also $a^2 + b^2 + c^2 + d^2 = 1$. It is also important to note that S^2 is not a Lie group.

Proposition. S^0 , S^1 , and S^3 are the only spheres that are Lie groups.

Another example to note in the three-dimensional group is the Heisenberg group, H , which is the group of 3×3 real matrices of the form

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}$$

We can make the following observation about the Heisenberg group:

Proposition. The Heisenberg group is a nilpotent group.

Proof. By an elementary argument, the center of H is the collection of matrices of the form

$$\begin{pmatrix} 1 & 0 & y \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Then we find that matrices in the quotient $H/Z(H)$ are of the form

$$\begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}$$

with a, c fixed. Then we find that $Z(H/Z(H)) = H/Z(H)$, so modding $H/Z(H)$ by its center gives the trivial group. Thus H is nilpotent. $\square$

In higher dimensions, there are other notable Lie groups; however, we will not delve into them much here. To mention complex simple Lie groups, the so-called “classical groups” consist of $SL_n(\mathbb{C})$ (for $n \geq 1$), $O_n(\mathbb{C})$ (for $n \geq 3$), and $Sp_{2n}(\mathbb{C})$ (for $n \geq 1$).

2 Lie Groups & Lie Algebras

Say G is a Lie group. The Lie algebra of G is the tangent space of G at its identity element e . However, we want an additional “structure” on the Lie algebra to capture some “information” about the Lie group (we will investigate what this means precisely shortly). First we discuss 1st order differential operators. On a smooth manifold, these have the form $\sum f_i(x_1, \dots, x_n) \partial / \partial x_i$. Vector fields on a smooth manifold M simply associate, to each point $p \in M$, a vector $V(p)$. If $M = \mathbb{R}^n$, then the vector fields gives a 1st order differential operator, since we can take the derivative of the vector (at each point) in each of the n directions given by the basis vectors of $\mathbb{R}^n$. Vector fields then give an easy way of viewing “infinitesimal automorphisms” of a smooth manifold M , since for each point $p \in M$, we can send it to the tip of the vector $V(p)$ (it is an automorphism in the sense that we can send each $p \in M$ to $-V(p)$).

Now we examine 1st order differential operators. We want to “combine” two 1st order differential operators and get another one. By a quick calculation, it may be verified that simply taking the product of two 1st order differential operators yields a 2nd order differential operator (via the product rule for derivatives). However, if we take the *commutator* of two 1st order operators, we get another one. Precisely, if D, E are 1st order differential operators, then the commutator given by $DE - ED$ is another 1st order differential operator. This is the additional “structure” we want for the Lie algebra. In this context, we define the *Lie bracket* of differential operators D, E to be $[D, E] := DE - ED$. (By applying the product rule repeatedly, it follows that if D, E have order m, n (respectively), then the bracket $[D, E]$ has order $m + n - 1$.) We define the *Lie algebra* L (of a Lie group G) to be the tangent space over the reals (or the complex numbers) at the identity of G so that the following two properties hold: (1): $[D, E] = -[E, D]$ and (2): $[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0$.

the latter of which is called the Jacobi identity.

We now investigate how to calculate the Lie algebra of a Lie group via a few examples. Say $G = GL_n(\mathbb{R})$; the identity element is just the $n \times n$ identity matrix. Say γ is a smooth path in G with $\gamma(0) = I$. Then we can write a Taylor expansion of $\gamma(t)$ as: $\gamma(t) = I + A_1 \cdot t + A_2 \cdot t^2 + O(t^3)$. Since the Lie algebra is a tangent space of G (which yields a first order approximation), then since we get no relation on a matrix in G , thus the Lie algebra of G can be any matrix, so $Lie(G) = M_n(\mathbb{R})$ (where $Lie(G)$ denotes the Lie algebra of G). Now say $G = SL_n(\mathbb{R})$, which consists of determinant 1 matrices. Then via the exponential map, matrices in $T_e G$ (this notation refers to the tangent space at the identity of G) are matrices X so that $\exp(tX) \in G$. We know $\det(e^{tX}) = 1$ by choice of G . Then we use the identity $\det(e^A) = e^{\text{trace}(A)}$. So $e^{\text{trace}(tX)} = 1$, so $\text{trace}(tX) = 0$, so $\text{trace}(X) = 0$, so we conclude that the Lie algebra of G consists of matrices X of trace 0. Now, we take $G = O_n(\mathbb{R}) = \{\text{matrices } A \mid AA^t = I\}$. Via first order approximation, we see that $A \approx 1 + \epsilon X$, so substituting into the defining relation for A , we get $(1 + \epsilon X)(1 + \epsilon X)^t = 1 + O(\epsilon^2)$. Expanding yields that it is equal to $1 + \epsilon X + \epsilon X^t + O(\epsilon^2)$, so we get $X + X^t = 0$. Thus the Lie algebra of $O_n(\mathbb{R})$ consists of matrices X so that $X + X^t = 0$.

We want to now investigate some fundamental questions about Lie groups & Lie algebras. Firstly, if G, H have isomorphic Lie algebras, then is $G \cong H$? The answer turns out to be no. G has the same Lie algebra as its connected component. Suppose now G is connected. Then any normal discrete subgroup of G is in the center, $Z(G)$. To see this, say Γ is a normal discrete subgroup of G and say $\gamma \in \Gamma$. Then look at the map $g \mapsto g^{-1}\gamma^{-1}g\gamma$. The image of this map lies in Γ , since the product $g^{-1}\gamma^{-1}g \in \Gamma$ (since Γ is assumed to be normal) and $\gamma \in \Gamma$, so by closure the image of the map lies in Γ . So we get a map $G \rightarrow \Gamma$, where G is connected and Γ is discrete. Thus the image is a point, and so it follows immediately that any normal subgroup of G lies in the center.

Now suppose we have a homomorphism of Lie groups $G \rightarrow H$. Then we do get a map of Lie algebras $Lie(G) \rightarrow Lie(H)$. However, the converse is false. To see why, set $G = \mathbb{R}$ and $H = S^1$. Then we have a homomorphism $G \rightarrow H$. (Note that S^1 is not simply connected.) Then for the Lie algebras, we get an isomorphism $\mathbb{R} \rightarrow \mathbb{R}$. However, there is no homomorphism $S^1 \rightarrow \mathbb{R}$. However, if we add an assumption, then a homomorphism of Lie algebras gives a homomorphism of Lie groups:

Proposition. If G is simply connected, then $Lie(G) \rightarrow Lie(H)$ gives a homomorphism $G \rightarrow H$.

Now suppose L is contained in a Lie algebra $Lie(H)$. Is there a closed subgroup corresponding to L ? The answer is no. To see why, say $G = S^1 \times S^1$, the 2-torus $\mathbb{R}^2/\mathbb{Z}^2$. Then take L to be multiples of $(x\alpha, x\beta)$ for $x \in \mathbb{R}$ and α/β irrational. Then the subgroup of all points $(x\alpha, x\beta)$ is dense in G (not closed).

We also have that any Lie algebra comes from a Lie group. We show this for a simple

case. Say L is a Lie algebra with trivial center, where the center of L is $\{x \mid [x, y] = 0 \forall y\}$, where $[\cdot, \cdot]$ is the Lie bracket. Define $\rho : L \rightarrow \{\text{linear transformations of } L\}$ so that $L \ni A \mapsto \rho(A)$, where $\rho(A)$ takes $B \in L$ to $[A, B]$. It follows that ρ is a homomorphism and by applying the antisymmetry and Jacobi identity of the Lie algebra, elements $A, \dots$ are part of a Lie group.

In essence, Lie algebras roughly correspond to simply connected Lie groups. We now compare connected groups and simply connected groups. Suppose H is connected and G is the universal cover of H . Say e is a basepoint of H . Then G is the collection of homotopy classes of paths in H that start at e , where paths are of the form $\gamma : [0, 1] \rightarrow H$ so that $\gamma(0) = e$ (note that the homotopy classes require paths that fix both $\gamma(0)$ and $\gamma(1)$). Then universal cover of H is simply connected and is also a group; multiplication in the group (that is the universal cover) is defined in the sense for each $h \in H$ and a path γ that goes from e to h , we can take an element g and multiply it on the right by $\gamma(t)$ for $t \in [0, 1]$. This group is locally isomorphic to H . So we get a map $\Gamma \rightarrow \text{simply connected group} \xrightarrow{\text{onto}} H$, where in this case, $\Gamma = \pi_1(H)$, the fundamental group of H .

Now we consider an example. Take $H = GL_2(\mathbb{R})$. We want to find the number of components of G and its fundamental group. For each element $B \in G$, we have an *Iwasawa decomposition*, namely we can write B has a product KAN , where K is a 2×2 real orthogonal matrix, A is a diagonal matrix, and N is a nilpotent matrix with 1s on the diagonal. Note that the collection of all real 2×2 diagonal matrices is just isomorphic to $\mathbb{R} \oplus \mathbb{R}$. Then we have that $\mathbb{R} \oplus \mathbb{R}$ is contractible, since, for each $(x, y) \in \mathbb{R} \oplus \mathbb{R}$, we can define $\gamma : [0, 1] \rightarrow \mathbb{R} \oplus \mathbb{R}$ so that $t \mapsto ((1 - t)x, (1 - t)y)$; since γ is continuous and $\gamma(0) = (x, y)$ and $\gamma(1) = (0, 0)$, thus γ is a contraction. So the the collection of all matrices A is contractible, and similarly, the collection of all matrices N is contractible. Thus the problem reduces to finding the number of components and fundamental group of the the orthogonal group $O_2(\mathbb{R})$. The group $O_2(\mathbb{R})$ has 2 components, given by rotations and reflections. (We could also notice directly that $GL_2(\mathbb{R})$ has two components, given by matrices of positive determinant and matrices of negative determinant.) Then the fundamental group of $GL_2(\mathbb{R})$ is isomorphic to that of $SO_2(\mathbb{R})$, which is isomorphic to that of S^1 , which is just $\mathbb{Z}$.

3 The Exponential Map

The function $\exp$ gives a map from the Lie algebra to the Lie group. At least for matrix groups, we can define $\exp$ explicitly in terms the Taylor series, i.e. for a matrix A , we define:

$$\exp(A) = 1 + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$$

Regarding convergence of the Taylor series, we choose the norm $|\cdot|$ on $\mathbb{R}^n$ and say $|A|$ is given by $\frac{\sup |Av|}{|v|}$ for nonzero vectors v . Then, by the Cauchy-Schwarz inequality, it follows that the Taylor series for $\exp(A)$ is absolutely convergent. By arithmetic, it may be verified that $\exp(A + B) = \exp(A) \cdot \exp(B)$ if A, B commute and $\exp(\lambda A + \mu A) = \exp(\lambda A) \cdot \exp(\mu A)$ for real numbers λ, μ ; from the second for-

mula it follows immediately that the map $\lambda \mapsto \exp(\lambda A)$ is a homomorphism that takes $\mathbb{R}$ to $GL_n(\mathbb{R})$. For simplicity, we shall here restrict to the study of matrix groups.

Proposition. Any finite-dimensional Lie group is locally isomorphic to a matrix group.

The formulas for the exp map given in the above paragraph are obtained under the assumption that A and B commute. Suppose now A, B do not commute. Keeping in mind the Taylor expansions for $\exp(A)$ and $\exp(B)$, we deduce:

$$\exp(A + B) = 1 + (A + B) + \frac{A^2 + AB + BA + B^2}{2!} + \dots$$

and

$$\exp(A) \cdot \exp(B) = 1 + A + B + \frac{A^2}{2} + \frac{B^2}{2} + AB + \dots$$

Comparing these formulas gives that $\exp(A) \cdot \exp(B)$ is the exponential of $A + B + [A, B]/2 +$ (higher terms). (These higher terms are given by the Baker-Campbell-Hausdorff-Dynkin formula, which we shall discuss later.) Intuitively, we can obtain $\exp(A) \cdot \exp(B)$ by taking the exponential of $A + B$, plus some correction terms. We now look at the following example. Suppose $A \in M_2(\mathbb{R})$. We wish to find $\exp(A)$. By the Cayley-Hamilton theorem from linear algebra, we can write the characteristic polynomial in A , say $A^n + \star A^{n-1} + \dots + \star I = 0$, where the $\star$ are some real numbers, so $A^n = -\star A^{n-1} - \dots - \star I$ (in the dimension n case). For our example, $n = 2$ so we observe the equation $A^2 = \star A + \star I$ for some real numbers $\star$. It follows that we can write $\exp(A)$ in the form $\star A + \star I$. Suppose for simplicity A is diagonal, i.e. we can write

$$A = \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}$$

Then simply exponentiating this diagonal matrix gives

$$\exp(A) = \begin{pmatrix} e^\lambda & 0 \\ 0 & e^\mu \end{pmatrix}$$

which we can write in the form $xI + yA$, for some real numbers x, y . Then we get

$$\begin{pmatrix} e^\lambda & 0 \\ 0 & e^\mu \end{pmatrix} = x \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + y \begin{pmatrix} \lambda & 0 \\ 0 & \mu \end{pmatrix}$$

so we deduce $e^\lambda = x + \lambda y$ and $e^\mu = x + \mu y$, and so $x = \frac{\mu e^\lambda - \lambda e^\mu}{\mu - \lambda}$ and $y = \frac{e^\lambda - e^\mu}{\lambda - \mu}$. (In the case that A is not diagonalizable, we may make look at the trace and determinant maps evaluated at A and obtain something similar.)

We now give an example of why exp is not onto. Suppose $G = SL_2(\mathbb{R})$ and $L = \mathfrak{sl}_2(\mathbb{R})$ be its Lie algebra, which is the collection of trace zero 2×2 real matrices. For a diagonalizable operator in the Lie algebra (say with eigenvalues λ, μ) we have the condition $\lambda + \mu = 0$. Then exponentiating the operator yields diagonal values e^λ, e^μ .

In the case of $\mathfrak{sl}_2(\mathbb{R})$, the eigenvalues λ, μ must either be both real or both imaginary. Consequently, we have e^λ, e^μ are positive real numbers with product 1 or we have $e^{i\lambda}, e^{-i\lambda} \in \mathbb{C}$, that lie on the unit circle in the complex numbers. In either case, we deduce $\text{trace}(e^A) \geq -2$. Then we have that matrices with trace less than -2 , like

$$\begin{pmatrix} -1/2 & 0 \\ 0 & -2 \end{pmatrix}$$

in $SL_2(\mathbb{R})$ are not in the image of the exponential map, so thus the exponential map is not surjective.

Suppose A, B are real $n \times n$ matrices (A, B lie in some Lie algebra). Then we had previously that a formula relates $\exp(A) \cdot \exp(B)$ and $\exp(A + B)$. Namely, for some δ , we can write $\exp(A) \cdot \exp(B) = \exp(A + B + \delta)$. We now investigate what this δ is. This is given by the Baker-Campbell-Hausdorff-Dynkin (BCHD) formula, which says that δ is can be written purely in terms of the Lie bracket $[\cdot, \cdot]$. Explicitly, δ has the form $[A, B]/2 + [A, [A, B]]/12 + [B, [B, A]]/12 + \dots$. This sum does not converge in general.

We will describe the proof of the BCHD formula. Say R is a free associative algebra on x, y (for noncommutative polynomials). Then we can think of the completion $\hat{R}$ of R as noncommutative power series. We define a coproduct in $\hat{R}$ by $\Delta : R \rightarrow R \otimes R$ (a ring homomorphism) so that $R \ni r \mapsto r \otimes 1 + 1 \otimes r$. So we get a map $\hat{R} \rightarrow \hat{R} \otimes \hat{R}$. For a group G , we define a *Hopf algebra* to be the group algebra $\mathbb{R}[G]$ (which is an associative algebra) so that we have a standard product $\mathbb{R}[G] \otimes \mathbb{R}[G] \rightarrow \mathbb{R}[G]$ (given by the product, as an algebra) and a coproduct $\Delta : \mathbb{R}[G] \rightarrow \mathbb{R}[G] \otimes \mathbb{R}[G]$ satisfying the condition given in the previous sentence. For any Lie algebra, we can construct a Hopf algebra (called the *universal enveloping algebra*). Say an element $r \in R$ is *primitive* if $\Delta r = r \otimes 1 + 1 \otimes r$. As from before, we conclude that the primitive elements of the algebra form a Lie algebra. Suppose r, s are primitive. Then $\Delta[r, s] = \Delta r \Delta s - \Delta s \Delta r$ (by the standard definition of the Lie bracket), which is equal to $[r, s] \otimes 1 + 1 \otimes [r, s]$. The exponential map takes primitive elements r to “grouplike” elements g so that $\Delta g = g \otimes g$. Since $\Delta r = r \otimes 1 + 1 \otimes r$, we then get $\Delta \exp(r) = \exp(\Delta r)$, which equals $\exp(r \otimes 1 + 1 \otimes r)$, which then equals $\exp(r) \otimes \exp(r)$. Now we can give an outline of the proof of the BCHD formula. Say $x, y \in \hat{R}$ are primitive elements. Then $\exp(x), \exp(y)$ are grouplike in the sense they have the form A , for $\Delta A = A \otimes A$. Then taking the logarithm of $\exp(x) \cdot \exp(y)$ gives a primitive element. So $\exp(x) \cdot \exp(y) = \exp(x + y + \delta)$, and this δ is given by $\log(\exp(x) \cdot \exp(y))$. Lastly, it remains to show that any primitive element of $\hat{R}$ is in the completion of the Lie algebra generated by x, y , which we will do shortly.

4 Universal Enveloping Algebra

We will now discuss a little about the universal enveloping algebra $U(L)$ of a Lie algebra L . $U(L)$ is an associative algebra generated by L , and we get a map $L \rightarrow U(L)$ from the Lie algebra to the universal enveloping algebra so that for elements $a, b \in L$ (sent to $a, b \in U(L)$ respectively, by abuse of notation) the commutator $[a, b]$ in L

is sent to $ab - ba \in U(L)$. As an example, say L is an abelian Lie algebra, i.e. the bracket of any two elements in L is 0. Say L has a basis $a_1, \dots, a_n$. Then $U(L)$ is the polynomial algebra $\mathbb{R}[a_1, \dots, a_n]$, since brackets map to commutators. As another example, say L is a Lie algebra spanned by elements a, b so that $[a, b] = b$. Then the universal enveloping algebra $U(L)$ is generated by a, b so that $ab - ba = b$, so the map $L \rightarrow U(L)$ “preserves” generators.

Now we want to find upper/lower bounds for the *size* of $U(L)$, i.e. we want to show $U(L)$ has the same *size* as the polynomial algebra on n generators, where $n = \dim L$ as a vector space. We now calculate the upper bound. Say $a_1, \dots, a_n$ is a basis of L . By looking at the bracket operation, we say that $a_{i_1}, \dots, a_{i_k}$ for all integer sequences $i_1, i_2, \dots$ so that $i_1 \leq i_2 \leq \dots$. We can filter $U(L)$ by choosing subspaces $F_0 \subset F_1 \subset \dots$ in L so that F_k is spanned by all $a_{i_1}, a_{i_2}, \dots$ with at most k a_j ’s in them. From this we can form a graded algebra as follows. Say $S(L)$ is the (symmetric algebra) polynomial algebra generated from L , which is commutative. We get a surjective map $S(L) \rightarrow F_0 \oplus F_1/F_0 \oplus F_2/F_1 \oplus \dots$, with the codomain having the “same size” as the universal enveloping algebra of L (with the notion given by the restrictions on the basis vectors, as seen earlier in this paragraph). The codomain admits a grading, so we get $F_i F_j \subset F_{i+j}$. Also we have that F_i and F_j commute up to terms in F_{i+j-1} . We now give the following theorem.

Theorem (Poincare-Birkhoff-Witt). The surjective map (as given in the above paragraph) is an isomorphism.

We now examine some applications of the above theorem. We first investigate how to recover the Lie algebra L from the universal enveloping algebra $U(L)$. Consider the following construction. Define the coproduct $\Delta : U(L) \rightarrow U(L) \otimes U(L)$ so that $\Delta(a) := a \otimes 1 + 1 \otimes a$. This gives a homomorphism of associative algebras.

Theorem. A Lie algebra L (for $\mathbb{R}$, or for any field of zero characteristic) consists of the primitive elements of the universal enveloping algebra $U(L)$.

Consider the following argument. Filter $U(L)$ as $F_0 \oplus F_1/F_0 \oplus F_2/F_1 \oplus \dots$ (as seen previously), which is a polynomial algebra. Suppose $f \in U(L)$ is primitive and $f \in F_n$ (by the grading of the filter). The filter is a universal enveloping algebra of a commutative algebra, as also seen previously. Then the image of f in F_n/F_{n-1} is primitive, so it is in F_{n-1} if $n > 1$, so $f \in F_1$, so $f \in L \subset F_1$. We have that L is the free Lie algebra on x, y . So $U(L)$ is the free associative algebra (which consists of noncommutative polynomials in the indeterminates x, y), so the primitive elements of $U(L)$ are just the elements of L .

5 Nonzero Characteristic

We now discuss the theory in characteristic $p > 0$. In previous cases, we considered the general linear groups, etc. over a field of 0 characteristic, such as $\mathbb{R}$ or $\mathbb{C}$. We look at the corresponding analogues over a field k with $\text{char } k > 0$. These analogues

$GL_n(k)$, etc. are called *algebraic groups*. In the complex case, we have groups $(\mathbb{C}, +)$ and $(\mathbb{C} \setminus \{0\}, \times)$. We can get a local isomorphism from the former to the latter by the exponential map given by $z \mapsto e^{2\pi iz}$, and we see that both of these groups have the same Lie algebra, $\mathbb{C}$. Now take a field k with $\text{char } k = p > 0$. The corresponding algebraic groups are $(k, +)$ and $(k^\times, \times)$. Although these have the same Lie algebra, k , they are not locally isomorphic; so by working over fields of nonzero characteristic, we lose some structure that we saw when working over fields of zero characteristic. $(k, +)$ is an example of a *unipotent group*, i.e. if we had a matrix representation of it, then all elements of the matrix group have eigenvalues equal to 1, so the matrices are upper-triangular with 1s on the diagonal. $(k^\times, \times)$ is an example of a *reductive algebraic group*, i.e. if we had a matrix representation of it, then all elements of the group are diagonalizable.

Restricting to nonzero characteristic fields yields also issues in the arithmetic. For instance, we do not have an exponential map in nonzero characteristic. This is because, looking at the Taylor series expansion for $\exp$, we have constants in the denominator and some of these are 0 in characteristic $p > 0$.

Over $\mathbb{R}$, in $U(L)$, L is the subspace of primitive elements v , where $\Delta v = v \otimes 1 + 1 \otimes v$. Recall $\Delta : U(L) \rightarrow U(L) \otimes U(L)$. However, in nonzero characteristic, this fails. Consider the Lie group $(k, +)$ with characteristic $p > 0$. Then $U(L)$ is a polynomial ring in D , spanned by $1, D, \dots, D^p, \dots$. (The space of differential operators are spanned by $1, D, D^{(2)}, \dots, D^{(p)}, \dots$. The primitive elements of $U(L)$ are of the form $D, D^p, D^{p^2}, D^{p^3}, \dots$. Then $\Delta D^k = \sum D^{(i)} \otimes D^{(k-i)}$. So the primitive elements are spanned by D . Since we are over nonzero characteristic, we get that 0-dimensional groups (which are the finite groups) of order p are connected, which is not the case over zero characteristic fields. Also, over nonzero characteristic, we have Lie algebras that do not correspond to Lie groups. To see this, consider the *Witt algebra*, which is a Lie algebra spanned by elements of the form $L_{i-1} := x^i \frac{d}{dx}$. Then

$$[L_m, L_n] = x^m \frac{d}{dx} x^n \frac{d}{dx} - x^n \frac{d}{dx} x^m \frac{d}{dx}$$

which is equal to $(n-m)x^{m+n-1} \frac{d}{dx} = (n-m)L_{m+n}$. By the definition of L_i , we see that the Witt algebra is an infinite-dimensional Lie algebra. Now look at characteristic $p > 0$, so for L_m , m is an element of $\mathbb{Z}/p\mathbb{Z}$ (for example). Then the bracket $[L_m, L_n]$ equals $(n-m)L_{m+n}$, so we get a finite-dimensional Lie algebra, which does not correspond to any simple algebraic group (by checking the classification of simple algebraic groups).

6 Bianchi Classification

The Bianchi classification is a classification of connected Lie groups/algebras for dimension at most 3. We aim to classify the Lie algebras, find the center of these Lie algebras, and see that the Lie groups are given by a simply connected group modulo the center. In dimension 0, the classification is trivial. In dimension 1, we have a Lie algebra $\mathbb{R}$, with the bracket given by $[\cdot, \cdot] = 0$, as $\mathbb{R}$ is abelian. Note also that

$(\mathbb{R}, +)$ is simply connected with $\mathbb{R}/\mathbb{Z} \cong S^1$. In dimension 2, we have a Lie algebra spanned by a, b with $[a, b] = \star$, for some value $\star$. For instance, we may take either $[a, b] = b$ or $[a, b] = 0$. So the possibilities for the dimension 2 case are the Lie algebra $\mathbb{R}^2$ (with the trivial bracket) or the Lie algebra of the $ax + b$ group (as discussed earlier). So the corresponding Lie groups are $\mathbb{R}^2/0$ (which is just $\mathbb{R}^2$), $\mathbb{R}^2/\mathbb{Z}$, $\mathbb{R}^2/\mathbb{Z}^2$, and the group generated by real matrices (with columns $\begin{pmatrix} a & 0 \\ b & 1 \end{pmatrix}$) with $a > 0$. So, we get a total of 4 Lie groups of dimension 2, up to isomorphism.

We now examine the dimension 3 case. Say we have a Lie algebra L , with dimension 3 (over the reals). We would like to examine if L has a 2-dimensional normal subalgebra M . We define M to be a *normal subalgebra* if $[m, L] \subset M$ for $m \in M$. (This corresponds to the notion of a normal subgroup.) A 2-dimensional subalgebra M must either be $\mathbb{R}^2$ or the $ax + b$ -algebra (as seen in the above paragraph). So L is generated by x, M . We have that the map $m \mapsto [x, m]$ is linear on M . If M is the $ax + b$ -group, then this linear map is of the form $m \mapsto [m', m]$ for some $m' \in M$. By subtracting m' from x , we can assume $x = 0$. So we can assume $M = \mathbb{R}^2$ with $[\cdot, \cdot] = 0$ and x to be any linear map.

Now we examine the possibilities for the linear map on M (with respect to the $ax + b$ -group). If the eigenvalues (for the map) are the same (and equal 0), then we get the 2×2 matrix filled with zeroes and the 2×2 matrix with a 1 in the top-right corner and zeroes elsewhere. If the eigenvalues are the same (but are nonzero), then we get the 2×2 identity matrix and the 2×2 identity matrix with a 1 above the diagonal. Now suppose the eigenvalues are different. If one of the eigenvalues is 0, then we get matrices of the form

$$\begin{pmatrix} 0 & \\ & 1 \end{pmatrix}$$

If we have a real nonzero eigenvalue, then we get matrices

$$\begin{pmatrix} 1 & \\ & a \end{pmatrix}, \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$$

If the eigenvalues are complex, then we get

$$\begin{pmatrix} a + bi & \\ & a - bi \end{pmatrix}$$

(where $a = 0$ is a special case). The 2×2 zero matrix is denoted as “Type I”. The Lie algebra is $\mathbb{R}^3$ (abelian), and the Lie groups are $\mathbb{R}^3$, $\mathbb{R}^3/\mathbb{Z}$, $\mathbb{R}^3/\mathbb{Z}^2$, and $\mathbb{R}^3/\mathbb{Z}^3$. We denote the case (for the 2×2 zero matrix with a 1 above the diagonal) as “Type II”, and the Lie algebra is the Heiseberg algebra generated by 3×3 matrices with values to be chosen for the entries above the diagonal (and zeroes elsewhere). Here, the Lie group is generated by matrices of the form

$$\begin{pmatrix} 1 & \star & \star \\ 0 & 1 & \star \\ 0 & 0 & 0 \end{pmatrix}$$

with trivial center. For the matrix with a 1 in the bottom-right corner (and zeroes elsewhere), we denote this as “Type III”, and we get the Lie group $\mathbb{R} \times$ (the

$ax + b$ -group). We keep going to get up to and including “Type VII” (these 8 types correspond to Thurston’s 8 geometries).

Now we look at groups with no 2-dimensional normal subgroup. So we examine the simple Lie algebras (a *simple Lie algebra* is a nonabelian Lie algebra with no nonzero proper ideals, with ideals defined in the canonical manner). There exists one 3-dimensional complex Lie algebra that is simple, namely $\mathfrak{sl}_2(\mathbb{C})$, which is the collection of 2×2 complex matrices with trace 0. There exist 2 3-dimensional Lie algebras (over $\mathbb{R}$) that are simple that give $\mathfrak{sl}_2(\mathbb{C})$ (when tensored with $\mathbb{C}$), which are $\mathfrak{sl}_2(\mathbb{R})$ and $\mathfrak{su}_2(\mathbb{R})$. The corresponding Lie groups are $SL_2(\mathbb{R})$ and $SU_2(\mathbb{R}) \cong S^3$. We notice that $SL_2(\mathbb{R})$ is not simply connected, it has a center isomorphic to $\mathbb{Z}$, and by taking the universal cover $\widetilde{SL_2(\mathbb{R})}$, which has center isomorphic to $\mathbb{Z}$. However, $SU_2(\mathbb{R}) \cong S^3$ is simply connected (has center of order 2), and we get $S^3/(\mathbb{Z}/2\mathbb{Z}) \cong SO_3(\mathbb{R})$. Thus we get 2 Lie groups for the two 3-dimensional simple real Lie algebras.

7 Engel’s Theorem

We now want to connect the ideas of nilpotent groups with Lie algebras. Recall that a group G is *nilpotent* if we have the following chain $G = G_0$, $G_1 := G_0/Z(G_0)$, $G_2 := G_1/Z(G_1)$, etc., and for some n , we have G_n is the trivial group. We define a Lie algebra L to be *nilpotent* similarly, i.e. if we have the chain $L_0 = L$, $L_1 := L_0/Z(L_0)$, $L_2 := L_1/Z(L_1)$, etc., and for some n , we have that L_n is the trivial Lie algebra.

Theorem (Engel). Suppose L is a Lie algebra and $L \subset \mathfrak{gl}_n(\mathbb{R})$. Then if all elements of L are nilpotent, then L fixes (i.e. $lv = 0$ for all $l \in L$) some $v \in \mathbb{R}^n$, with $n > 0$ and $v \neq 0$.

(It must be noted that, however, $L \subset \mathfrak{gl}_n(\mathbb{R})$ can be nilpotent as a Lie algebra but not contain solely nilpotent matrices. To see this, we can consider the 1×1 (non-nilpotent) matrix (1) in $\mathfrak{gl}_n(\mathbb{R})$.) We now give a sketch of the proof of Engel’s theorem. The main idea is that we repeatedly use induction on the dimension of L . Firstly, we notice L acts on itself by nilpotent operators, in the sense we have a map $l : L \rightarrow L$ so that $m \mapsto [l, m]$, which we can think of as “left multiplication”. Clearly the product/sum of two nilpotent elements is nilpotent (see binomial expansion). So if an input $m \in L$ is nilpotent, then $l(m) = [l, m]$ is nilpotent. Next, suppose L is nonzero and contains a subalgebra of codimension 1, call it M , i.e. we have $[L, M] \subset M$. Pick M to be a maximal subalgebra not equal to L . We look at M acting on the quotient L/M . By induction, if we take $v \in L/M$, $v \neq 0$, and v is fixed by M , then we have $\mathbb{R}v \oplus M$ is a subalgebra that must equal L , as we assumed M is maximal. So $[v, M] = M$, so M is an ideal. Now suppose $L \supset M$, with M an ideal of codimension 1. Say V_0 is the set of vectors fixed by M . By induction, we see that V_0 is nontrivial. Then we have $L = \mathbb{R}x \oplus M$, and let x act on V_0 , so it fixes a nonzero vector of V_0 . So we conclude V_0 is fixed by L , and so we are done. $\square$

Theorem (Kolchien). If G is an algebraic group in $GL_n(k)$, then if G is nilpotent,

then G is conjugate to the subgroup of matrices of the form

$$\begin{pmatrix} 1 & & \star \\ & \ddots & \\ 0 & & 1 \end{pmatrix}$$

(where the matrices have all eigenvalues equal to 1).

Suppose now we have a Lie algebra L (over the real numbers or a field of zero characteristic) with elements matrices that are upper-triangular with zeroes elsewhere. (Thus L is nilpotent. However, the converse, that is, L is nilpotent implies L contains solely matrices that satisfy this condition, is false as seen previously.) Then the exponential map is an isomorphism from L to the corresponding Lie group. By nilpotence, the exponential map yields just a finite series. So, the set of matrices $\exp(a)$ (for $a \in L$) is a simply connected Lie group.

We now look at an example of a nilpotent Lie group that is not simply connected, and not a matrix group. Take G to be a quotient, in this case the Heisenberg group modulo the collection of matrices of the form

$$\begin{pmatrix} 1 & 0 & n \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

(with n in the integers). The “denominator” of this quotient G lies in the center $Z(G)$. Then the Lie algebra of G is spanned by the elements

$$x = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix},$$

$$z = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

(note that z spans the center of the Lie algebra). We get the relations $[x, y] = z$, $[z, x] = [z, y] = 0$. Now look at the image of z in $M_n(\mathbb{C})$. From the map $G \rightarrow GL_n(\mathbb{C})$, we get a corresponding map $Lie(G) \rightarrow M_n(\mathbb{C})$. Notice now that $\exp(z) = 1$, using the Taylor series of $\exp$. Thus z is diagonalizable. z is in the center of G also. So pick an eigenspace V of z in $\mathbb{C}^n$ and consider the action $G \curvearrowright V$. So, in V , z has the image

$$\begin{pmatrix} a & & 0 \\ & \ddots & \\ 0 & & a \end{pmatrix}$$

where $a \in \mathbb{C}$. From before we have $[x, y] = z$, so the trace of $[x, y]$ equals the trace of z , so the trace of xy minus the trace of yx equals 0, so the trace of z is zero. Note that $\text{trace}(z) = \dim(V) \cdot a$, so $a = 0$ as we are working over a field of zero characteristic. Thus z acts trivially on any eigenspace, so the action of z is trivial on V , so $G \rightarrow GL_n(\mathbb{C})$ is trivial on $\exp(z)$, and we are done.

8 Lie's Theorem

The key idea of Lie's Theorem is to connect the ideas of solvable groups with solvable Lie algebras. First, we begin with some definitions. Recall that a group G is *solvable* if we have a decreasing chain $G = G_0 \supset G_1 \supset \cdots \supset 1$ so that G_i/G_{i+1} is abelian with G_i a normal subgroup of G_{i-1} . Similarly, we say a Lie algebra L is *solvable* if $L = L_0 \supset L_1 \supset \cdots \supset 0$ is a chain of Lie algebras so that L_i/L_{i+1} is abelian with respect to the bracket.

Theorem (Lie). Any finite-dimensional complex representation of a solvable Lie algebra has an eigenvector $l(v) = \chi(l)v$ for some linear map $\chi : L \rightarrow \mathbb{C}$, where χ is viewed as an "eigenvalue".

We get a similar result for connected solvable Lie groups. However, the corresponding result fails for disconnected groups. It also fails for finite groups, since we can take the group $G = S_3$ (which acts on $\mathbb{R}^2$ or $\mathbb{C}^2$), which does not have any eigenvalues. It also fails for nonzero characteristic fields. For instance, we can take L to be the Lie algebra spanned by $1, x, d/dx$ acting on $\langle 1, x, x^2, \dots, x^{p-1} \rangle = k[x]/(x^p)$. Then L is a nilpotent Lie algebra and $\dim \langle 1, x, \dots, x^{p-1} \rangle = p$, with no eigenvectors.

Suppose $L \subset \mathfrak{gl}_n(\mathbb{R})$. Suppose L is solvable. Then L (in $GL_n(\mathbb{C})$) is conjugate to a subalgebra of the (solvable) algebra of upper-triangular matrices. So every solvable subalgebra is conjugate to a subalgebra of the algebra of upper-triangular matrices.

Corollary. If L is solvable, then $[L, L]$ is nilpotent.

Now we give an outline of the proof of Lie's Theorem. For case 1, suppose L is abelian and suppose L acts on a complex vector space V . So $L = \langle l \rangle \oplus M$ (with M having codimension 1). Then by induction, pick eigenspace W of M , where $W = \{mv = \chi(m)v \text{ for some } \chi : M \rightarrow \mathbb{C}\}$. l acts on W , because l commutes with M , so we can pick an eigenvector of l on W , and we are done. For case 2, suppose L is solvable. Then we can pick a codimension 1 ideal M of L (since L is solvable), and consequently, M is solvable. So $L = \langle l, M \rangle$. Then pick an eigenspace W of M (by induction). So $mw = \chi(m) \cdot w$, where $\chi : M \rightarrow \mathbb{C}$. If l maps W to W , then we can finish by picking an eigenvector of l on W . If $lm = ml$ when restricted to w , then it's easy to check $l : W \rightarrow W$. So, we want to show $[l, M] = 0$ on W iff $\chi([l, M]) = 0$. We have $L = \langle l, M \rangle$ acts on V and contains W , where W is an eigenspace of M , so $mw = \chi(m) \cdot w$. We want to show now $\chi([l, m]) = 0$. Choose $w \in W$. Look at $w, lw, l^2w, \dots, l^kw$, where $l^{k+1}w \in \langle w, lw, \dots, l^kw \rangle$. Inductively define $W_1 := \langle w \rangle$,

$W_2 := \langle w, lw \rangle$, and so on. So by construction $W_1 \subset W_2 \subset W_3 \subset \dots \subset W$. If $v \in W_i$, then $mw = \chi(m)v$ modulo W_{i-1} (by definition of the W_i 's), since $ml^i v = lml^{i-1}v + [m, l]l^{i-1}v$ (where the first term is $l\chi(m)l^{i-1}v + lW_{i-2}$, and the second term lies in W_{i-1}). So, on W_k , m looks like the $k \times k$ matrix

$$\begin{pmatrix} \chi(m) & & \star \\ & \ddots & \\ 0 & & \chi(m) \end{pmatrix}$$

So, the trace of m (restricted to W_k) is $k\chi(m)$ (for $m \in M$). Now change $m \mapsto [l, m]$. Then the trace of $[m, l]$ (restricted to W_k) is $k\chi([l, m])$ (the k is nonzero as we are zero characteristic). The left-hand side is equal to zero, so thus $\chi([l, m]) = 0$, and we are done.

9 Integration

We now want a way to integrate over Lie groups. In calculus, the “nice” cases for integration consist of integrating over a compact subset of the reals. Here, the “nice” analogue is integrating over compact Lie groups. We have that G has a *left-invariant* measure μ , i.e. a measure μ satisfies $\mu(U) = \mu(gU)$ (for all $g \in G$), where $U \subset G$ is an open subset of G . Similarly, a *right-invariant* measure is μ so that $\mu(U) = \mu(Ug)$ for all $g \in G$. As an example, we have the standard Lebesgue measure on $G = \mathbb{R}^n$.

Now we give three methods of showing the existence of a measure on a Lie group. For the first method, we utilize the result that any locally compact group has a Haar measure (which is left-invariant). As an application, we have that the finite-dimensional representations of compact (Lie) groups are completely reducible. For if we have an exact sequence $0 \rightarrow A \rightarrow B \rightarrow C \rightarrow 0$, then it is split, i.e. $B = A \oplus C$. For instance, if we say G is the group of 2×2 upper-triangular real matrices with 1's on the diagonal, then if we consider its action on $\mathbb{R}^2$, then we get a split exact sequence $0 \rightarrow \mathbb{R} \rightarrow \mathbb{R}^2 \rightarrow \mathbb{R} \rightarrow 0$.

For the second method, we can use n -forms on G . As from real analysis, n -forms correspond roughly to measures (we say n , where $n = \dim G$). So, given a function f that takes in an input $(x_1, \dots, x_n) \in G$, we can integrate the expression $f(x_1, \dots, x_n)dx^1 \cdots dx^n$ (using the Einstein convention). However, as seen in the theory of differentiable manifolds, we must consider the orientation when integrating. To take care of this, we choose an orientation on G (there are 2 orientations if G is “nice”). Suppose for instance $G = \mathbb{R}$. Then the map $x \mapsto -x$ (and correspondingly $dx \mapsto -dx$) takes Lebesgue measure $\rightarrow$ Lebesgue measure.

For the third method, we give an explicit n -form on G . We look now at some examples. If $G = \mathbb{R}^n$, then, as we have seen before, we take the measure to be $dx^1 \cdots dx^n$. If G is discrete, we can take the counting measure (as coming from the discrete topology on G). For $G = (\mathbb{R}_{>0}, \times)$, we take the measure to be $\frac{dx}{x}$. Then for a positive real number a , we have $\frac{dax}{ax} = \frac{dx}{x}$, so left-invariance holds. (As an example,

from complex analysis we have the gamma function defined by $\Gamma(s) = \int_0^\infty e^{-t} t^{s-1} dt$, which can be written as $\int_0^\infty e^{-t} t^s \frac{dt}{t}$. From the second expression, we have the measure $\frac{dt}{t}$.

We now look at another example. Say G is the group of matrices of the form

$$\begin{pmatrix} x & y \\ 0 & 1 \end{pmatrix}$$

with x, y real and $x > 0$. Clearly this group is isomorphic to the group of real pairs (x, y) with x positive. Induced from matrix multiplication, the group law we get on the pairs is given by $(a, b)(x, y) = (ax, ay + b)$. The measure we are looking for is $\frac{dx dy}{a^2}$ (the denominator comes from taking the absolute value of the determinant of the Jacobian matrix). This measure is left-invariant by a simple calculation. If we instead wanted a right-invariant measure, we can define look at right multiplication on the pairs, given by $(x, y) \mapsto (ax, bx + y)$. Then the absolute value of the determinant of the Jacobian matrix is a , so the desired measure is $\frac{dx dy}{a}$.

Now take $G = \mathbb{C}^\times$, the group of nonzero complex numbers under multiplication. The map $z \mapsto zz_1$ multiplies volumes by $|z|^2$, and the measure we get is $\frac{dx dy}{x^2 + y^2}$, where $z = x + iy$.

Lastly, we consider the following example. Let G be the group of matrices of the form

$$\begin{pmatrix} x_{11} & x_{12} & x_{13} \\ 0 & x_{22} & x_{23} \\ 0 & 0 & x_{33} \end{pmatrix}$$

with x_{ii} positive for all i . We show what the left-invariant and right-invariant measures on G are. For the left-invariant measure, we claim it is

$$\frac{\prod_{i \leq j} dx_{ij}}{|x_{11}|^3 \cdot |x_{22}|^2 \cdot |x_{33}|^1}$$

To do this, left multiply $X \in G$ by $A \in G$ to form Y , i.e. $Y := AX$. If we show the measure on Y is the same as that for X , we are done. Write X, A as in the definition of G . Then for

$$Y = \begin{pmatrix} y_{11} & y_{12} & y_{13} \\ 0 & y_{22} & y_{23} \\ 0 & 0 & y_{33} \end{pmatrix}$$

the multiplication $Y = AX$ implies the equations $y_{11} = a_{11}x_{11}$, $y_{12} = a_{11}x_{12} + a_{12}x_{22}$, $y_{13} = a_{11}x_{13} + a_{12}x_{23} + a_{13}x_{33}$, $y_{22} = a_{22}x_{22}$, $y_{23} = a_{22}x_{23} + a_{23}x_{33}$, and $y_{33} = a_{33}x_{33}$. Now we calculate the Jacobian matrix J of the functions y_{ij} in terms of the variables x_{ij} (each y_{ij} gives a row and each x_{ij} gives a column). Finding the partial derivatives

$\frac{\partial y_{ij}}{\partial x_{lk}}$ (for $i \leq j$ and $l \leq k$) gives

$$J = \begin{pmatrix} a_{11} & 0 & 0 & 0 & 0 & 0 \\ 0 & a_{11} & 0 & a_{12} & 0 & 0 \\ 0 & 0 & a_{11} & 0 & a_{12} & a_{13} \\ 0 & 0 & 0 & a_{22} & 0 & 0 \\ 0 & 0 & 0 & 0 & a_{22} & a_{23} \\ 0 & 0 & 0 & 0 & 0 & a_{33} \end{pmatrix}$$

So we get $|\det J| = |a_{11}|^3 \cdot |a_{22}|^2 \cdot |a_{11}|^1$. Then we have $dY = |\det J|dX = |a_{11}|^3 \cdot |a_{22}|^2 \cdot |a_{33}|^1 dX$. By the claimed measure, we have that the measure on Y is

$$\frac{dY}{|y_{11}|^3 \cdot |y_{22}|^2 \cdot |y_{11}|^1}$$

which equals

$$\frac{dX \cdot |a_{11}|^3 \cdot |a_{22}|^2 \cdot |a_{11}|^1}{|a_{11}x_{11}|^3 \cdot |a_{22}x_{22}|^2 \cdot |a_{33}x_{33}|^1}$$

which just equals the measure on X . So the measure on AX equals that on X , so the claimed measure is left-invariant. Now we show that the measure given (on X) by

$$dX = \frac{\prod_{i \leq j} dx_{ij}}{|x_{11}|^1 \cdot |x_{22}|^2 \cdot |x_{33}|^3}$$

is the right-invariant measure. By a similar process, say $Y := XA$ (so we right-multiply X by $A \in G$). Analogously, we get a Jacobian matrix

$$J = \begin{pmatrix} a_{11} & 0 & 0 & 0 & 0 & 0 \\ a_{12} & a_{22} & 0 & 0 & 0 & 0 \\ a_{13} & a_{23} & a_{33} & 0 & 0 & 0 \\ 0 & 0 & 0 & a_{22} & 0 & 0 \\ 0 & 0 & 0 & a_{23} & a_{33} & 0 \\ 0 & 0 & 0 & 0 & 0 & a_{33} \end{pmatrix}$$

So $|\det J| = |a_{11}|^1 \cdot |a_{22}|^2 \cdot |a_{33}|^3$ (from which we can guess what the right-invariant measure is). Then $dY = |\det J|dX$, so the measure on Y is

$$\frac{dY}{|y_{11}|^1 \cdot |y_{22}|^2 \cdot |y_{33}|^3}$$

and by substituting in the equations for the y_{ij} in terms of the x_{ij} and cancelling, we get that it's equal to

$$\frac{dX}{|x_{11}|^1 \cdot |x_{22}|^2 \cdot |x_{33}|^3}$$

and so we have shown the claimed measure is right-invariant. This group G is thus an example of a group where the left and right-invariant measures are not equal.